TYPOGRAPHIC SYSTEM FOR STRUCTURING PHONETIC ALPHABETS

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Abstract

This project investigates how the book can function as a tool for structuring and comparing phonetic alphabets from the 15th century until recent works. It explores how macro-typographical systems can make sense of complex linguistic material and create conditions for comparison and future research. The goal is to provide a clear and accessible overview of phonetic alphabets: how and for wat language they are designed, which speech sounds they incorporate, and how their graphemes function as visual forms. By developing a structured framework anchored in book design, that focuses on important information and comparative potential, the project tries to visualize both the differences and commonalities between phonetic alphabets across a wide range of them. It is a new way of presenting these types of alphabets by connecting them to their respective informative and formal features, which allows for comparing and drawing conclusions like common design strategies, visual connections between individual graphemes, grapheme functionality, the influence of historical context...

The framework materializes as a series of three books: the Encyclopedia of Phonetic Alphabets (EOPA), the Catalogue of Vowels (COV), and the Catalogue of Consonants (COC). The encyclopedia and catalogues approach the same content from a different angle—the EOPA offering contextual, historical and conceptual insights into 51 phonetic alphabets, spanning from 1440 up until the present, combining it with high-quality visuals, while the COV and COC organize the individual graphemes by phoneme—thus their speech sound—, enabling visual comparison of their designs. As a series, the books are designed to not only present information, but to trigger cross-comparing inside and between books. Their shared design and structure create a system that enhances comparison between volumes.

The thesis concludes with a method for working with phonetic alphabets through linguistic and design perspectives. It highlights how the act of designing and organizing this material through the physical medium of the book can reveal connections that might otherwise stay hidden. In doing so, it gives opportunities for further research into recurring graphic strategies, formal relations between phonemes, and the visual concepts of phonetic alphabets.

Content Table

Glossary

The glossary is placed at the beginning of the thesis to provide readers with immediate access to the terminology used throughout the work. Since many of these terms are context-related and may not be familiar to a general audience, presenting them upfront ensures clarity and accessibility.

Phonetics - A branch within linguistics where they study human speech sounds.

Phonetic alphabet - A set of characters designed to represent human speech sounds from any language. They function as an aid for pronouncing while reading and a medium for phonetic transcription.

International Phonetic Alphabet or IPA - The phonetic alphabet that is the current standard for phonetic transcription. Example: The quick brown fox jumps over the lazy dog. / ðə kwɪk bɹaʊn fɑks ʤʌmps ˈoʊvɚ ðə ˈleɪzi dɔg.

Writing system - A structured method of visually representing spoken language through symbols. Examples: alphabetic, abjad, logographic, syllabaries, ...

Script - The set of symbols used within a writing system. Example: alphabetic – Latin, Cyrillic, Greek, ...

Latin script – Originally developed by the Romans, influenced by the Etruscan and Greek scripts and became the basis for a wide range of modern scripts. Today, it is the most widely used script in the world as it supports languages like English, French, Spanish... The Latin script is used in this thesis. **(I1)**

Cyrillic script – The script used by many Slavic languages like Russian, Bulgarian, Serbian... **(I2)**

Alphabet - The fixed set of symbols taken from a script, which is linked to a language. Some languages use the same alphabet; others also incorporate other symbols of the same script. Example: alphabetic – Latin – Dutch alphabet

Orthography - The set of rules and principles determining how a language is written. This includes spelling, grammar, punctuation, and capitalization.

Convention - An agreement or rule accepted by the users of a language or writing system.

Grapheme - The smallest functioning unit in a writing system that can distinguish meaning between words. They are the visual representation of human speech sounds or phonemes. Example: in ‘word’ the ‘w’, ‘o’, ‘r’ and ‘d’ are all separate graphemes.

Phoneme - The smallest functioning unit of sound in spoken language that can distinguish meaning between words. Example: the sound from the ‘o’ in ‘word’ is a phoneme.

Speech sound - A physical sound produced by the vocal cords of a human.

Consonant - A speech sound produced with a constriction or closure in the vocal tract. The sound is created by stopping the flow of the air in the mouth.

Vowel - A speech sound produced with an open vocal tract. The sound is produced by letting the air escape the mouth in an easy fashion.

Monophthong - A singular vowel sound, which can be defined as long or short depending on the duration of the airflow. Example: the ‘o’ in ‘word’ or the ‘oo’ in ‘broom’.

Diphthong - A combination of two monophthongs. Example: the ‘y’ in ‘sky’, which is transcribed as ‘aɪ’.

Triphthong - A combination of three monophthongs. Example: the ‘i’ in ‘fire’, which is transcribed as ‘aɪə’

Speech organs - The anatomical parts used for producing speech sounds. These include the lungs for airflow, vocal cords for trembling (voicing), tongue, lips, teeth, and mouth.

IPA-parameters - The parameters used by the IPA for classifying speech sounds.

In the case of monophthongs **(I3)**:
Raising of the tongue - closed/near-closed/close-mid/mid/open-mid/near-open/open
Part of the tongue being raised – front/central/back
Shape of the lips – rounded/unrounded
Example: the ‘i’ in ‘win’ is a ‘near-close front unrounded vowel’.

In the case of consonants:

Three main consonant divisions:

Pulmonic consonants – the air comes from the lungs
Co-articulated consonants – two simultaneous places of articulation
Non-pulmonic consonants – the air does not come from the lungs

Three parameters **(I4)**:

Voicing or trembling – voiced/unvoiced
Place of articulation or where in the mouth the sound is produced – bilabial/labiodental/linguolabial/dental/alveolar/postalveolar/retroflex/palatal/velar/uvular/pharyngeal/glottal **(I5)**
Manner of articulation or how the air escapes the mouth – nasal/plosive/(non-)sibilant affricate/(non-)sibilant fricative/approximant/tap/trill/lateral affricate/lateral fricative/lateral approximant/lateral tap

Example: the ‘k’ in ‘book’ is an ‘unvoiced velar plosive’.

Transparent and non-transparent language - The regularity of grapheme-phoneme mapping. If the number of graphemes is equal to the number of phonemes, that language is deemed transparent. If the number of graphemes is lower, it means that some graphemes will have multiple phonemes, which makes the language opaque or non-transparent.

Auxiliary language–A language, which is often constructed, that is used to communicate when two people speak different native languages.

State of the Art

This project is the result of a fascination for typography and its utilitarian means of visualizing the world's different languages. Through my studies in Graphic Design and Typography and traveling in free time, several languages surfaced in my life like English, French, Portuguese, Maltese, Italian, Spanish, Polish, etc... All these languages sound different and have different vocabularies, although all use the Latin script in their written form. Using the same script for multiple languages means using the same graphemes for various phonemes, which means that each grapheme doesn't always carry the same phoneme or possibly represents more than one phoneme. This aspect of written language determines what is called the 'transparency of a language'. Finnish is the most transparent language in the world while English is the most opaque. (Halmari, H. (1998). *Transparency versus opaqueness: The case of Finnish and English*) For this issue of unpredictability of phonemes and the goal of helping with pronunciation, phonetic alphabets were invented. These alphabets are based on the sounds of spoken language in order to accommodate pronunciation without the difficulties of the unpredictability of opaque languages. Each grapheme of the language dealt with has one fixed phoneme. The idea of a phonetic alphabet traces back to the 15th century and has seen a variety of experiments and proposals.

Former research largely focuses on the efficacy of phonetic alphabets in educational settings (Dumanggas, C. D. et al. 2024 & Ehri, L. C. et al. 2001) and mostly uses the International Phonetic Alphabet (IPA), the current standard for transcribing speech sounds, as its subject (Dumanggas, C. D. et al. 2024 & Setiyono, M. S. et al. 2019). This makes the ongoing research from a design perspective very limited and thus gives limited insights into the design of the phonetic alphabets themselves. The alphabets are viewed separately instead of relating subjects, which means there is little coverage of the possible connections between them on a formal and informational level. The lack of a systematic approach to organizing knowledge on phoneme-grapheme pairings, the systems behind it, and the grapheme design seems to further limit progress in this field and asks for in-depth exploration.

The project aims to be this exploration by covering the alphabets on an informational and visual basis. In order to do so it turns to two well-established book genres: the encyclopedia and the catalogue. The encyclopedia offers, by definition, a systematic overview that combines textual explanation with visual representation, making it the ideal medium for contextualizing and classifying phonetic alphabets. Meanwhile the catalogue is a genre traditionally used to display and compare objects in a consistent format, making it the perfect platform for organizing the graphemes. By combining both genres, the project creates a complimentary series: the encyclopedia provides the contextual background of the alphabets, while the catalogues enable detailed visual comparisons. Together, they represent the ambition of the project of structuring and organizing the phonetic alphabets for future design and linguistic research. They enable the reader to draw conclusions like common design strategies, visual connections between graphemes, historical influences... Similar projects exist like the "The Missing Scripts Project" (ANRT, IDG & SEI. (2019). *The Missing Scripts Project*) **(I6)**, a joined effort by Atelier National de Recherche Typographique (ANRT), Institute Designlabor Gutenberg (IDG) and Script Encoding Initiative. Their goal is to create a digital library for every script of the world and try to recover forgotten ones. Another example is "Scriptsource" (SIL International. (n.d.). *ScriptSource: The world's writing systems*) **(I7)**, an initiative by SIL Global, created in order to share their knowledge of scripts and to provide a platform for people to share their knowledge. Their desired outcome is for every script to be understood and supported by software developers and industry standards. Lastly there is "Omniglot" (Ager, S. (1998). *Omniglot*) **(I8)**, an independent digital platform by Simon Ager, made for learning about languages, alphabets, other writing systems, phrases and numbers. Works related in terms of design are the "Octavo Series" (Atelier Carvalho Bernau. (2011). *Octavo Series*) **(I9)** by Atelier Carvalho Bernau where all the books in the series are linked through macro typographical parameters. The goal was to show that good design does not have to be more expensive in terms of materials or added physical elements and to incorporate simple layout choices to accommodate complex information, by adding color coding or consciously employing typographic hierarchy.

Problem Statement

Regarding a structured overview of phonetic alphabets and providing a platform for other designers, most of the projects mentioned are similar in their intent, but they lack innovation in some ways. "The Missing Scripts Project" (ANRT, IDG & SEI. (2019). *The Missing Scripts Project*) **(I6)** is equal in goal and considers every script or alphabet but is limited in execution (for now). They do provide contextual information on the script but do this by means of links to other websites, hereby sacrificing cohesiveness and a clear overview of information. The information on these other sites is mostly from a narrative perspective, also lacking structure and therefore not addressing different kinds of reading methods which are prevailing easily searching and finding information for different contexts (Bessemans, A. (2024, November 29). *De rol van typografie in een multi-divers leeslandschap (lezing)*. Congres Buitenkant van het boek, Den Haag, Nederland / Bessemans, A. *Designforchung als Basis für gure Gestaltung*. 2024. in Sieghart, S. & Lorbach, R. P. Gutes Design für Leichte Sprache. Praxis und Theorie zur DIN SPEC 33429. Verlag Julius Klinkhardt, Bad Heilbronn 2024.). Although they show one exemplary grapheme of each script, a full set of symbols is not always present thus lacking on a visual basis. They do make up for redirecting the user to a Unicode page of the scripts of which there are graphemes. "Scriptsource" (SIL International. (n.d.). *ScriptSource: The world's writing systems*) **(I7)** could be a digital counterpart to this project as it works in a structured manner and highlights aspects such as history and use, the graphemes, even going into design and typography, which fonts support the graphemes and an image of each grapheme. A flaw could be the fact that they largely depend on public additions, which makes most information incomplete or non-existent and doubtful in their accuracy. "Omniglot" (Ager, S. (1998). *Omniglot*) **(I8)** provides detailed information on constructed, fictional or traditional writing systems with a database of over 2100 case studies. It gives context and how the scripts work, along with an image of the script or alphabet. While the type of information, especially how the script works, is one of the strong points of this website, it is given in one or a couple paragraphs in a narrative way. This makes finding specific information less easy as it is one cluster of text, and they do not alter the layout other than better fit the appropriate reading method. The imagery is pixelated most of the time which also could interfere with visual research. Although they do include phonetic alphabets, it is not their main focus. This means that the linguistic side isn't always provided, meaning the grapheme-phoneme mapping. As this is an important aspect of a phonetic alphabet it should be included when discussed. As mentioned, previous research mostly focuses on the efficacy of the alphabet (Ehri, L. C. et al. (2001); Setiyono, M. S. et al. (2019); Dumanggas, C. et al. (2024)). This means they solely look at how 'good' the alphabet works for learning pronunciation or learning a language. Typographic aspects or contextualization are excluded, while the design could have an influence on the functionality of such an alphabet. The most used case study is also the IPA, making these types of research very limited in scope and quantity. The International Phonetic Alphabet or IPA (Passy, P. et al. (1886). *International Phonetic Alphabet*) **(A17)** is the standardized phonetic alphabet for phonetic notation, as it represents all sounds in human language. It is used in the field of linguistics, but also by singers and actors. It provides a one-to-one grapheme-phoneme relationship and has every human speech sound included. It also allows for a detailed and accurate classification of phonemes and provides the needed nuances of accents and subtle differences in sounds.

This MA-project aims to combine the positive sides and to improve the shortcomings of the projects discussed above. First of all, all information and imagery of the alphabets will be contained in one encyclopedia as there will be no need for digital redirections or additional sources. An encyclopedia was chosen for its trait of giving visuals combined with textual information; in this case showing the alphabet and giving context to it. This physical copy enables the reader to immediately compare the phonetic alphabets—going for a quantitative approach—on a visual level and to find needed information in a fast way through macro-typographical elements (thus typographic hierarchy) and fixed layouts. Additionally, there will be two catalogues with a focus on the individual graphemes rather than the alphabets, purely focusing on the visual aspects and giving an extra tool for comparing and studying them. These three books are linked to each other through macro typographical features and visual cues, giving the opportunity for immediate cross-comparison between the books. The physicality of the artefacts shows the alphabet in a printed environment and enables the reader to add notes or drawings of themselves.

Objectives

This research is a systematic review of phonetic alphabets in order to provide a clear framework for linguists and designers through which they can investigate them in a structured manner. The framework will be translated into an encyclopedia—combining textual information, which provides context, with visuals; covering the many aspects of a phonetic alphabet—and catalogues—showing only imagery which enables visual comparative research, providing essentially a complete list of every phoneme. This overview of experimentation between typography and language aims to be an important source of information and inspiration for linguists and typographic design. Primarily an answer will be sought for the research question: 'How can a 'book-as-a-tool' framework be developed to compare and classify static phonetic alphabets?'.

To achieve these goals, we must start with collecting a broad range of case studies—phonetic alphabets—and researching them. To accommodate the linguistic side research on the sound classification system of the IPA is needed, with this the graphemes can be indexed by their respective phoneme. To have a framework the criteria must be determined—what is important information? —and these parts will be taken into the design of the books. When the typographical elements are determined, the content simply needs to be put in.

Methodology

Collecting & Researching

As mentioned, the first step is to collect several case studies. This was done through digital research, acquiring information on the mentioned websites, Wikipedia and the original publications of the alphabets, which are books or manuscripts containing the guide for the alphabet **(I10)**. Information was scarce and most of the time incomplete, so every source available was taken into consideration. Imagery was mostly of low quality, so the source images were manually traced in Adobe Illustrator. This created the opportunity to use the graphemes on any scale possible and to show visuals of the highest quality. As it was not clear if copyright on these publications and images were present and if so, still valid, these images will only be utilized for academic purposes.

Sound Classification

To index the individual graphemes and to provide the linguistic side of the project, an understanding of the sound classification system of the IPA was needed. A clear summary and explanation were found on Wikipedia as it provides simple charts and explanations for each parameter, with this information taken from the IPA itself. Within this system there are two groups: vowels and consonants. For vowels there are no subgroups and thus only one chart **(I11)**. This chart has three parameters relating to the speech organs. A vowel can be classified by the shape of the lips, the raising of the tongue and the part of the tongue being raised. For consonants however, there are three subgroups: pulmonic consonants, co-articulated consonants and non-pulmonic consonants. Each subgroup has their own chart **(I12, I13 and I14)**, but they share the same parameters: the voicing, place of articulation and manner of articulation. As said this system was used to index the individual graphemes. But as there was no prior knowledge of linguistics, software was used to classify the sounds, namely ChatGPT. In most publications a sample word was given for each phoneme of the alphabet, for example 'the 'a' in 'man''. The prompt used with ChatGPT was as followed: "Can you determine the vowel/consonant sound for 'x' in the English (/other language) word 'xxxx'. For an additional security layer of indexing, the findings of ChatGPT were sometimes checked with an IPA transcription site.

Contextual Aspects

For the framework we need to determine the important pieces of information. To choose them the following questions need to be taken into account: While everything is relevant, which is the essential information of the alphabet? Does it provide context for the alphabet and is it needed? Could it have influenced the design of the graphemes? After going through all the information collected and keeping these questions in mind, the following framework parts were determined:

The inventor–it is important to know who made the alphabet as it reveals the perspective the alphabet was made with. Were they solely a linguist, a designer or something totally different? Their expertise has an influence on the graphemes as their knowledge can vary. As well as their nationality. Their name, profession and eventual notable features were taken as content.

The time period–The time when the alphabet was created significantly impacts the design. Not only the time spirit, but limitations of technology or overall knowledge probably had effects on the alphabet. For the time period the year of creation is given along with the socio-cultural context of that time, if it was deemed related to the alphabet itself. A clear example could be the Deseret Alphabet, which has a clear link to the immigration in America during that time as it was created to introduce new members into the church, and to learn them English.

The purpose of the alphabet–Why the alphabet was made is one of the key factors of the design. A phonetic alphabet purely for transcription won't necessarily have notable features, while for example Alexander Melville Bell's "Visible Speech" (Bell, A. M. (1867). *Visible Speech or Physiological Alphabet*) **(A11)** was made for deaf people and was designed to show what happens with the speech organs during articulation.

The use and lifespan of the alphabet–If and how the alphabet was used, gives insights if the alphabet's speech visualizations can work. It can also show the results of certain design choices, positive or negative. The project translates this into information if the alphabet was used, for what and if it was known, for how long.

The intended coverage–While talking about language and phonetic alphabet, it is important to know for which languages they were meant. This could again influence the design as, for example, the combination of languages using the Latin **(I1)** and Cyrillic script **(I2)** could trigger graphemes combined of both. It is also interesting to see the geographical reach of each alphabet. This parameter is either indexed as the specific language or noted as 'various' if it was a group of languages. Auxiliary means that it was an auxiliary language. The geographical aspect is indicated on a map, as a visual representation.

The technology of reproduction–As said with the time period, technology might be a limiting factor; for how it was made, but also for its reproduction. Handwritten alphabets give more freedom in design, but might trigger difficulties with reproduction, while it could be the contrary for mechanical typesetting. There are three categories: MANUAL, MECHANICAL or DIGITAL.

The number of graphemes–This number indicates the complexity of a phonetic alphabet, but also its reach. If the number is very high, it is likely that the alphabet incorporates a lot more language than an alphabet which has a low number. The number of graphemes also affects the learning curve of the alphabet.

The case of the alphabet–This holds a direct relationship to the design as it potentially obliges the inventor to design alternate graphemes for upper and lowercase graphemes. Capitals also add readability as they add another layer of meaning to words or sentences, so leaving them out might introduce some difficulties with reading. The case is indicated by 'I' or 'II' meaning one or two cases or as UNICAMERAL or BICAMERAL.

The typography of the alphabet (I15)–The typography in this case is defined as the type of graphemes used in the alphabet. There are three categories created: EXISTING—graphemes which already exist in other traditional orthographies—, DERIVED—graphemes which are derived from these existing graphemes—and NEW—graphemes which have no apparent link to existing graphemes and are newly created for the alphabet. Balancing between familiarity and novelty is one of the issues when making a phonetic alphabet. Tracking this balance through time shows how designers navigated through this problem and how it affected their work.

The concept of the alphabet–Inspired by the purpose, this determines the design. After analysis four categories were created: UNSYSTEMATIC—no logical system behind the design of the graphemes other than convention or aesthetics—, SYSTEMATIC—the presence of a logical system, but this is not based on any research—, THEORETICAL—a system which is based on research, meaning papers or theoretical foundations—and VOCAL CORDS—trying to visualize what happens with the speech organs during articulation.

The typography and concept of the alphabet are somewhat linked to each other. An alphabet based on the vocal cords will (almost) always have newly created graphemes. Most of the time an unsystematic alphabet will have existing or derived graphemes, but this doesn't necessarily exclude the use of new graphemes.

Encyclopedia of Phonetic Alphabets/EOPA

For the encyclopedia we consider mainly the framework, as it holds the textual information on the alphabets, which then combined with visuals gives a clear view of each alphabet. The purpose of the encyclopedia, and the project, is to provide information in a structured and easy navigational way. Thus, this publication offers all textual information included in the framework along with every visual piece of content collected. The title of this book is "The Encyclopedia of Phonetic Alphabets". **(I16)**

Content Table

The content table gives an overview of the collected phonetic alphabets, listed in chronological order. For every alphabet the technical aspects are presented that can be defined by categories or numbers, which give a first and concise description of the alphabet, followed by the page number **(I17)**:

- **The name of the alphabet**–The name
- **The typography of the alphabet** - Color coding was used as the three categories could be mixed, which is easily visualized through color. Existing graphemes is dark green, derived graphemes is lighter green and new graphemes is bordeaux-like.
- **The case of the alphabet**–I or II cases
- **The technology of reproduction** - Manual, mechanical or digital
- **The number of graphemes**
- **The design concept** - Unsystematic, systematic, theoretical or vocal cords
- **The intended coverage**
- **The year of creation**

Entries

Each alphabet has its own chapter with a fixed layout, which is the same for every alphabet. The first spread is the title spread **(I18)**, with on the left page an exemplary grapheme of the alphabet and on the right the name of the alphabet in its own transcription. The color of the pages represents the color coding of the alphabet.

The second spread **(I19)** holds the textual information with all parts of the framework present. On the top left there are the elements mentioned in the content table with the name, case, technology of reproduction, number of graphemes, design concept and intended coverage. The left page has 5 paragraphs from left to right and top to bottom: information on the inventor, the time period when it was created, the purpose of the alphabet, explanation on the concept and the use and lifespan. It is meant to be read in this way, unless the reader is searching for specific information, as every paragraph adds another layer on the previous one until you have to whole picture. The paragraphs are small and relatively thin so the text can be read in a fast, but understandable fashion and contain only the necessary information. Next to the time period paragraph is a picture of the inventor. Beneath the text is a Gall-Peters projection map where the intended geographical coverage is mapped, along with the country of origin of the alphabet. This projection was chosen for its neutral representation of the world in terms of geographical area, which better highlights the geographical range of the alphabet. On the right page there is the sample phrase, which is the same for every chapter: “The quick brown fox jumped over the lazy dog. He did not have to, but he thought it was the cheeky thing to do.” This is partly the standard phrase used for checking and presenting Latin script typefaces as it holds every letter of the Latin alphabet. A second phrase was added to provide a completer picture of the alphabet in action and to have some additional phonemes, in English at least. On each side of the spread is the color coding presented in color blocks. By placing the blocks on the side, an effect on the exterior of the book was achieved **(I20)**. In this way the information is not only limited to the inside of the book and also gives hints on the outside.

The third and final spread **(I21)** shows the entire set of graphemes. The graphemes are indexed by their respective phoneme and the color of their typography **(I22)**. Here the reader can see how the graphemes relate or don’t relate to each other and gives them the opportunity to compare them with other alphabets in the book. The elements of the content table are once again in the top left corner of the spread. With a high number of graphemes, it is possible a fourth or fifth spread needs to be added to the chapter. With a collection of 51 alphabets and over 3700 graphemes the EOPA resulted in a publication of 336 pages.

Digital layer

For research and design purposes, a digital layer was added to the book project in the form of a QR code **(I23)**. This QR code is put in the front of the book and directs to a digital library, which contains all the digital visual representations of the alphabets, the graphemes and audio files of each phoneme, if it was available. The library provides the user with the opportunity to use the alphabets, if copyright is not implemented, and to link the actual speech sound to the grapheme. While a classification might give some sense, hearing the actual sound does extend another layer of understanding.

Technical Attributes

The size of the book closed is B5—176 by 250mm—to make it compact, but big enough to accommodate the content. This size makes it easier and ergonomical to navigate through the book, using the thumb and index finger. The top margin of the pages is 30mm while the rest are 13mm. The 30mm gives room for the elements of the content table and provides space between this and the paragraphs. The grid which exists within the margins is 3 columns x 4 rows with a white space of 4mm between them. This structures the 5 paragraphs, profile picture and map in a clear and simple way. The typeface used is Helvetica for its neutral look and timeless aesthetic. The goal was to have no distractions from or interferences with the phonetic alphabet, which is why every additional image, and the geographical map are in grey values. The colors used for the color coding have no significant meaning and are inspired by Hanzo Wada’s “Dictionary of Color Combinations. Volume I”.

Catalogue of Vowels/COV & Catalogue of Consonants/COC

While the EOPA combines text with imagery, the COV and COC solely focusses on the visuals of the alphabet—the graphemes—with little addition of text. The framework has less influence, while the sound classification system and charts determine the design. The goal of the catalogues is to give designers the opportunity to compare graphemes within a specific phoneme, rather than comparing between alphabets. The COV **(I24)** contains every monophthong and diphthong—different vowel sounds—grapheme while the COC **(I25)** covers every pulmonic, co-articulated and non-pulmonic consonant grapheme.

Content Tables

COC

The monophthongs and diphthongs each have their own content table and color, both being shades of blue. **(I26 and I27)** The content table of the monophthongs **(I26)** is a translation of the IPA chart for vowel classification. On the top side are markers indexed by the parameter of the part of the tongue being raised. On the sides are markers indexed by the parameters of the raising of the tongue. Like with the chart **(I11)**, you can visualize how the tongue moves/looks during the articulation of a certain vowel sound. For the third parameter—shape of the lips—the shapes are either rounded or angular, indicating respectively a rounded or unrounded vowel. The numbers inside the shapes are the page numbers of the respective phonemes. For example, the close front unrounded vowel graphemes can be found on page 12, going on until page 17, where on page 18 the close front rounded vowel graphemes begin. The color contrasts between the shapes serve as a heatmap, showing the relative popularity of the vowel sounds throughout all alphabets.

The content table of the diphthongs **(I27)** is not a translation, since there is no IPA chart for diphthongs. Here it was chosen to use the IPA graphemes for the indexing of the markers. The top markers indicate the first grapheme in the combination and the side markers indicate the second grapheme. The content table functions the same as the one of the monophthongs with the page numbers and color contrasts.

COV

The pulmonic, co-articulated and non-pulmonic consonants each have their own content table, which are all translations of their respective IPA chart, and color which are shades of orange/red. **(I28, I29 and I30)** It works the same as the content table of monophthongs with the page numbers and color contrasts as heat map. The only thing different are the parameters. The top markers are indexed by the parameter of place of articulation and the side markers are indexed by the parameter of manner of articulation. For the third parameter of voicing the shapes are either a square or a plus-shape, indicating respectively an unvoiced or voiced consonant. For example, the unvoiced bilabial nasal consonant graphemes can be found on page 14, going on until page 15, where on page 16 the voiced bilabial nasal consonant graphemes begin.

Panels

Each book is divided into their subgroups—monophthongs and diphthong for the COV and pulmonic, co-articulated and non-pulmonic consonants for the COC—as they also have their own content table, with each phoneme having their own panel. Only the subgroups have a title spread, where just the subgroup is mentioned, while the panels run continuously throughout the book without interruption from title pages.

Each panel has the same fixed layout. **(I31 and I32)** On the top and sides, the respective markers of the phoneme are highlighted and in the top left corner that name of the phoneme is mentioned. The same with the EOPA, the markers are put over the edge of the paper, uncovering information on the outside of the book **(I33)**. As the manner of articulation is the parameter that is handled first in the book, the exterior of the book shows the popularity of this parameter. The same grid of the EOPA is used in both catalogues, enabling the books to show 24 graphemes per spread. For each grapheme minimal information is added **(I31 and I32)**, which links back to their alphabet and the EOPA, being the name of the alphabet, technology of reproduction, design concept, intended coverage and color coding of the alphabet. In horizontal direction over the spread, each grapheme stands on the same invisible baseline, showing their relative position and size to each other. For the COV this resulted in a collection of 54 vowel sounds and 1042 graphemes, while the COC has 213 consonant sounds and 2294 graphemes, making the books respectively 148 and 392 pages.

This does not add up to the over 3700 graphemes presented in the EOPA. This is because a part of those graphemes represented accentuated sounds, meaning they were nasalized, palatized, rhotic... While this adds another layer of specificity to the sound classification, it was not necessary for this project and leaves room for an expert in linguistics to study and classify these graphemes.

Digital layer

The same QR code from the EOPA **(I23)** is provided in both catalogues. The audio files of the phonemes are indexed by their respective page number in the COV or COC. This makes tracking the sound you are looking at in the book easier and creates an extra connection to the publications.

Technical Attributes

The COV and COC both use the same grid, size and typeface as the EOPA. By making no changes to these design aspects, all three books are linked to each other through these subconscious connections.

Conclusion

This project ends in a comprehensive and structured trilogy of publications: the Encyclopedia of Phonetic Alphabets (EOPA), the Catalogue of Vowels (COV), and the Catalogue of Consonants (COC). Together, this series forms a physical, typographic research tool that presents the different experiments and works in which graphemes and phonemes are mapped to each other.

The EOPA provides a framework for categorizing phonetic alphabets and an overview of 51 phonetic alphabets, combining contextual and linguistic data with high-quality visuals. The presentation of each alphabet is done through a consistent layout system that focuses on clarity, navigation and comparability, which are supported by macro typographical features such as color coding, fixed layouts and informational layers. The color coding helps with navigation through the book, giving the first piece of information and enables macro navigation, where the grid provides the structural consistency throughout the entries. The margins, normally used for notes, now contain information units with additional data and provide a contextual layer. The text and visuals are the comparative layer, which include the content and information to study and understand the alphabet. This hierarchical system of informational layers goes from fast scanning to a deeper engagement, if the reader finds the aspects he or she was looking for. It gives the opportunity for quick navigation, the making of comparisons between alphabets and to see the alphabets in a tactile format.

The COV and COC shift the focus from entire alphabets to individual graphemes, organizing them by phoneme through the IPA classification system. The catalogues offer pure visuals which enable the comparison between graphemes within a specific human speech sound, highlighting the differences and similarities on a formal level.

The tactile format of the publications allows for instant cross-comparison within and across the volumes, offering an experience that digital platforms like “Scriptsource” or “Omniglot” can’t replicate. Beyond their value as reference tools, this series opens up new possibilities for typographic and linguistic research. Designers could use the books to identify recurring design strategies, understand how form follows function across contexts and explore the balance between familiarity and innovation in grapheme creation. Linguists could examine how the visual side of these alphabets may affect usability. This structured archive makes a path for expansion of the collection and the research, particularly in the unexplored areas of accentuated sounds and potentially multiple variations of one phonetic alphabet. The project not only documents phonetic alphabets but places them into a visual system preparing them for research, going an extra step in relation to the original objective. Through tactility and methodical organization, it offers a powerful platform for future inquiry across linguistics and graphic design. During the research other aspects came to light which add another layer to the project, such as the digital library with the high-resolution images and even the catalogues, which were rather a by-product than a starting idea. This shows that the research is flexible, but also responsive to the data which was collected. Like with the books, new insights led to (design) choices which strengthened the project. This series of books creates a platform where phonetic alphabets are seen through their formal aspects and while tactility is an advantage in relation to the other projects, there are possibilities of creating a digital medium for the encyclopedia and catalogues. Taking the same framework and structure as a base, it gives the opportunity to incorporate animated or moving alphabets in the collection. It also enables the collection of alphabets to expand, whereas with printed works there is a need for multiple publications. I am interested in how this could unfold, and I may take this project into my next study, which will be centered around digital environments and website design.